

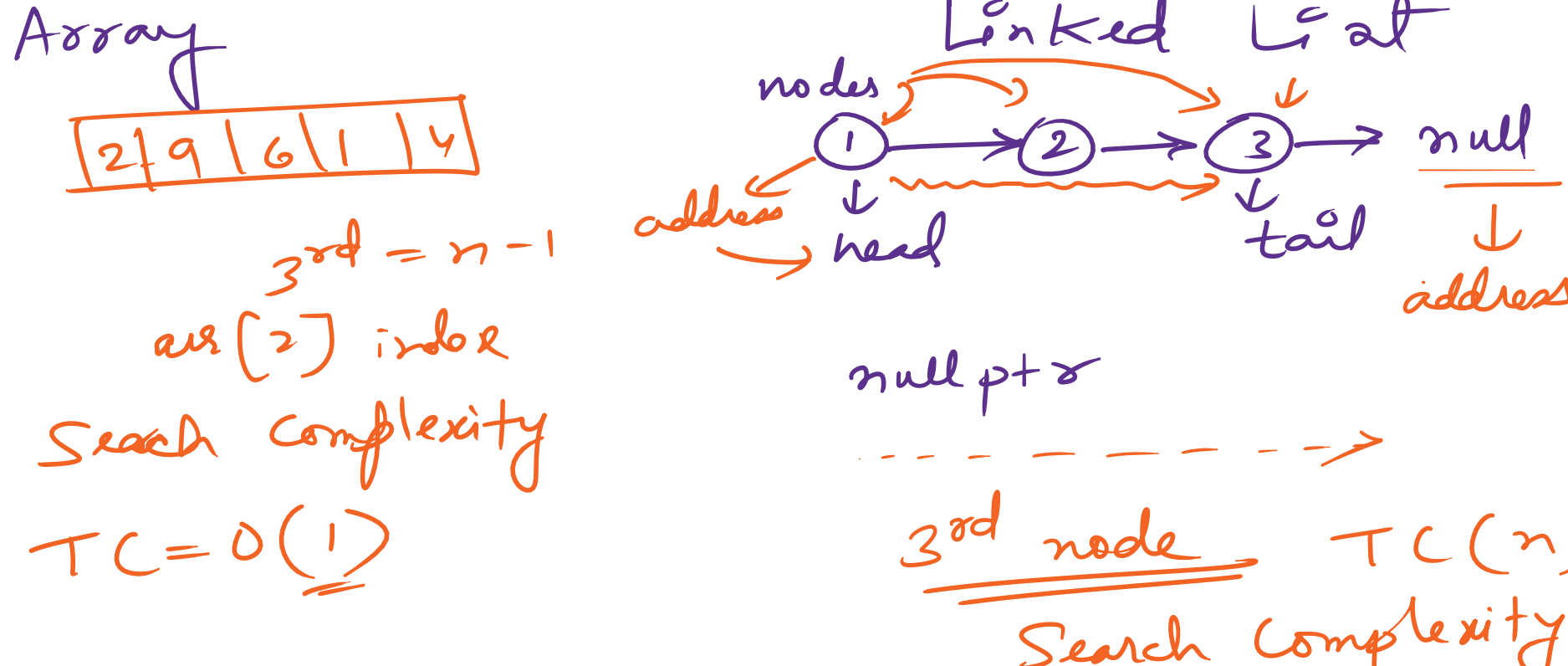
Linear data Structures : →

- ① Stacks, Queues, Arrays
- ② Linked Lists
 - ↳ Singly → Forward Traversal
 - ↳ Doubly → Both Side "
 - ↳ Circular

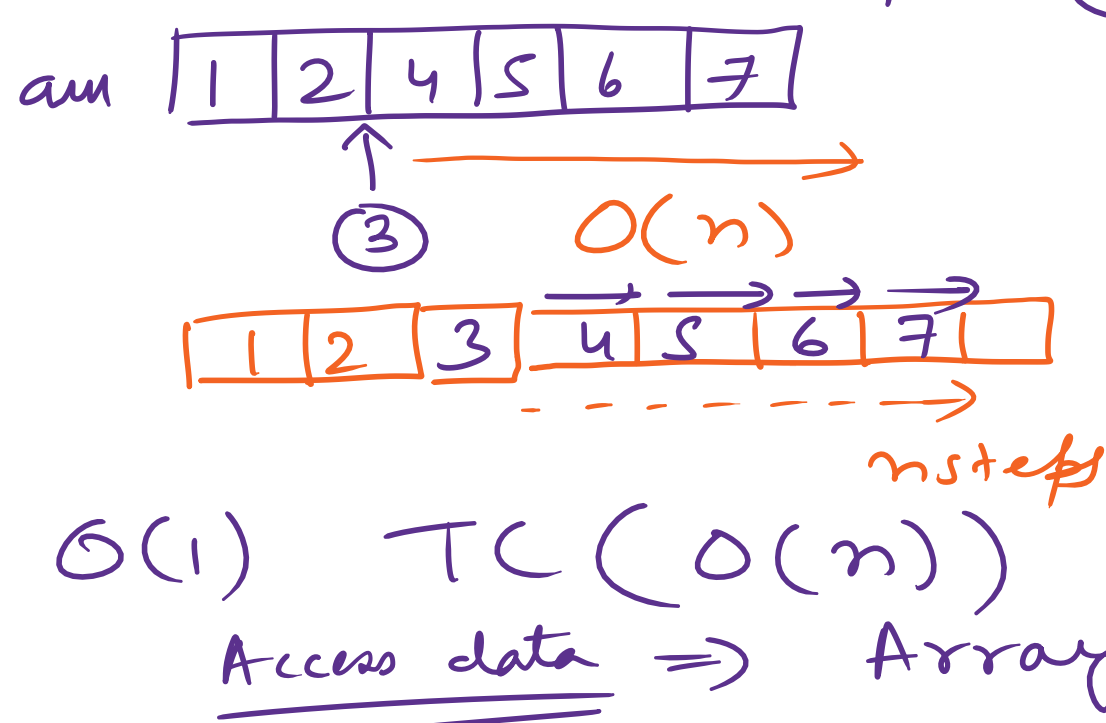
Array → Index

12	9	1	3
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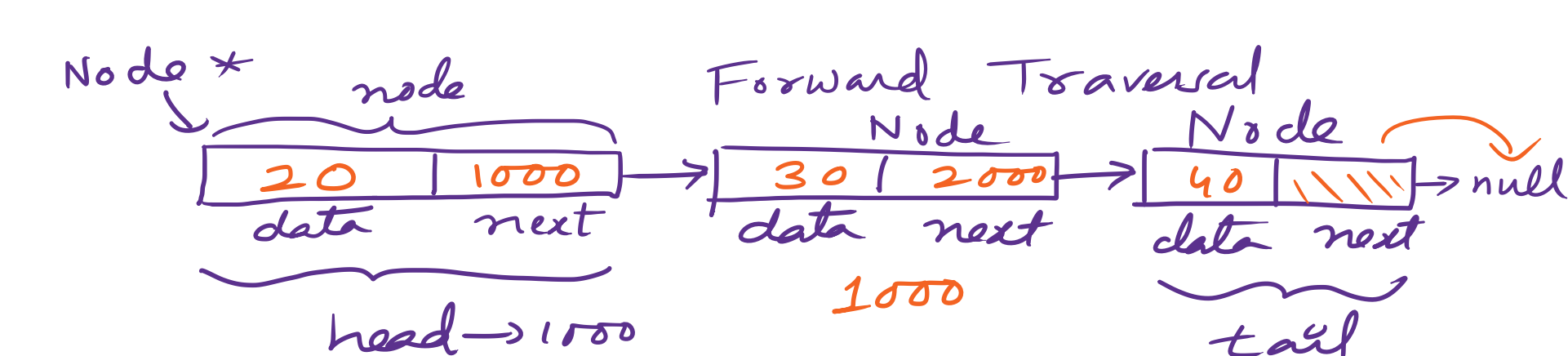
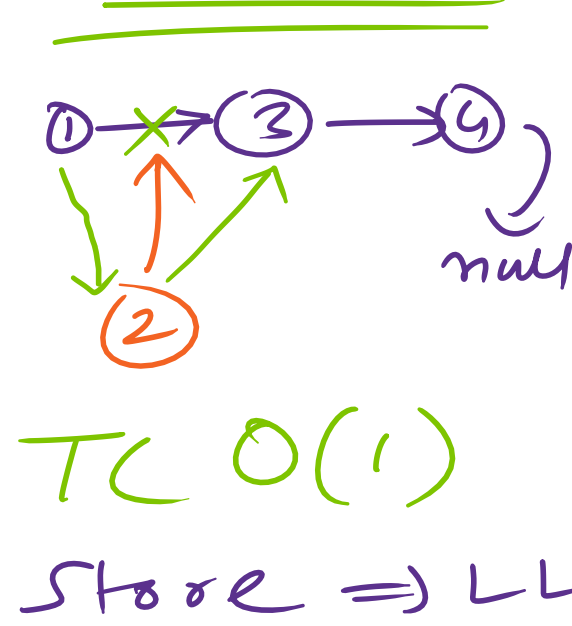
 $arr[3] \rightarrow TC \rightarrow O(1)$
 $(n-1) \rightarrow \text{index}$



Insertion Time Complexity :



Linked List



✓ struct Node {
 public int data;
 Node * next;
 }
 ✓ class Node {
 private int data;
 Node * next;
 }

Operations in a linked list :

- ① Insert →
 - ① Insert at Head
 - ② Insert at Tail
 - ③ Insert after Specific
- ② Deletion →
 - ① Delete Head
 - ② Delete Tail
 - ③ Delete Specific
- ③ Display $1 \rightarrow 2 \rightarrow 3 \rightarrow \text{null}$
 $node \rightarrow node \rightarrow node$
 $1 \rightarrow 2 \rightarrow 3$

Insert At Head : $1 \rightarrow 2 \rightarrow 3 \rightarrow \text{null}$
 $head \rightarrow 1$ $second \rightarrow 2$ $tail \rightarrow 3 \rightarrow \text{null}$
 $iah(head, nd)$

② $1 \rightarrow 2 \rightarrow 3 \rightarrow \text{null}$
 $h \rightarrow \text{pointer } **$
 $Node * node = \text{new Node}();$
 $node \rightarrow data = \text{newdata};$
 $node \rightarrow next = h$
 $h = node$
 $newdata$ ①
 $head = 1$

Insert At Tail : Create a new Node
 $new Node \rightarrow next = \text{null}$

① Empty : $head = \text{nullptr}$
 make the new node as head

② Only one node : $1 \rightarrow \text{null}$
 $\text{if}(head \rightarrow next == \text{null})$
 $h \rightarrow next = \text{new Node}$
 $l = l \rightarrow next$

③ Multiple nodes : $1 \rightarrow 2 \rightarrow 3 \rightarrow \text{null}$
 $l = h$
 $\text{while}(l \rightarrow next != \text{null})$

$l \rightarrow next = \text{new Node}$
 $l = h$
 $\text{while}(last \rightarrow next != \text{null})$

Steps : Head :

- ① Create Node → newdata
- ② Add data to the node
- ③ Point next of new node → head
- ④ Point head to new node



$h \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow \text{null}$
 $h \rightarrow 1$
 $new Node \rightarrow 1$
 $new Node \rightarrow next = 2$
 $h \rightarrow next = 1$
 $h = 1$
 $if (prev == \text{null})$
 $cout << "cannot";$
 $return;$
 $new Node \rightarrow next = p \rightarrow next$
 $p \rightarrow next =$
 $new Node$
 $new Node \rightarrow 1$
 $new Node \rightarrow next = 2$
 $h \rightarrow next = 1$
 $h = 1$

$1 \rightarrow 2 \rightarrow 3 \rightarrow 4$
 insert after target
 pointer (head, 1)